We say that they have the form of a number. But values cannot be assigned to them in a way that is consistent with the usual rules of addition and mutiplication of numbers. We call these expressions indeterminate forms. Although they are not numbers, these indeterminate forms play a useful role in the limiting behaviour of a function.

John (Johann) Bernoulli discovered a rule using derivatives to compute the limits of fractions whose numerators and denominators both approach zero or ∞. The rule is known today as l'Hôpital's Rule (pronounced as Lho pi tal Rule), named after Guillaume de l'Hospital's, a French nobleman who wrote the earliest introductory differential calculus text, where the rule first appeared in print.

7.5.2 The l'Hôpital's Rule

Suppose $f(x)$ and $g(x)$ are differentiable functions and $g'(x) \neq 0$ with

$$\lim_{x\to a} f(x) = 0 = \lim_{x\to a} g(x) \text{. Then } \lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

$$\lim_{x\to a} f(x) = \pm\infty = \lim_{x\to a} g(x) \text{. Then } \lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

7.5.3 Indeterminate forms $\frac{0}{0}, \frac{\infty}{\infty}, 0\times\infty, \infty-\infty$

Example 7.33

Evaluate : $\lim_{x\to 1}\left(\frac{x^2-3x+2}{x^2-4x+3}\right)$.

Solution

If we put directly $x=1$ we observe that the given function is in an indeterminate form $\frac{0}{0}$. As the numerator and the denominator functions are polynomials of degree 2 they both are differentiable. Hence, by an application of the l'Hôpital Rule, we get

$$\lim_{x\to 1}\left(\frac{x^2-3x+2}{x^2-4x+3}\right) = \lim_{x\to 1}\left(\frac{2x-3}{2x-4}\right)$$
$$= \frac{1}{2}.$$

Note that this limit may also be evaluated through the factorization of the numerator and denominator as $\frac{x^2-3x+2}{x^2-4x+3} = \frac{(x-1)(x-2)}{(x-1)(x-3)}$.

Example 7.34

Compute the limit $\lim_{x\to a}\left(\frac{x^n-a^n}{x-a}\right)$.

Solution

If we put directly $x=a$ we observe that the given function is in an indeterminate form $\frac{0}{0}$. As the numerator and the denominator functions are polynomials they both are differentiable.

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